

CivicPulse

AI-Powered Municipal Grievance Redressal & Smart City Platform

Track: Smart Cities / AI & Automation

Target Audience: Municipal Corporations & Citizens

Tech Stack: HTML5, Node.js, Gemini Vision AI, Leaflet

1. Problem Statement Understanding

Core Challenge: Urban infrastructure management suffers from inefficient citizen reporting, slow manual triage by municipal departments, and a complete lack of public transparency.

For Citizens

- **High Friction:** Filing a complaint requires navigating complex portals with up to 20 tedious input fields.
- **Opaque Status:** Citizens receive zero visibility into ticket progress after submission, causing frustration.
- **High Abandonment:** Over 65% of minor infrastructure issues go unreported due to portal complexity.

For Municipal Authorities

- **Manual Triage Bottlenecks:** Staff spend hours reading vague, miscategorized complaint descriptions.
- **Duplicate Overhead:** Multiple reports for the same incident (e.g., a major pothole) clog the ticket system.
- **Lack of Spatial Intelligence:** Absence of real-time heatmaps prevents proactive maintenance dispatching.

2. Proposed Solution

Solution Concept: CivicPulse introduces a zero-form, multimodal AI platform that converts a single photo into a fully categorized, geotagged, and severity-assessed civic ticket instantly.

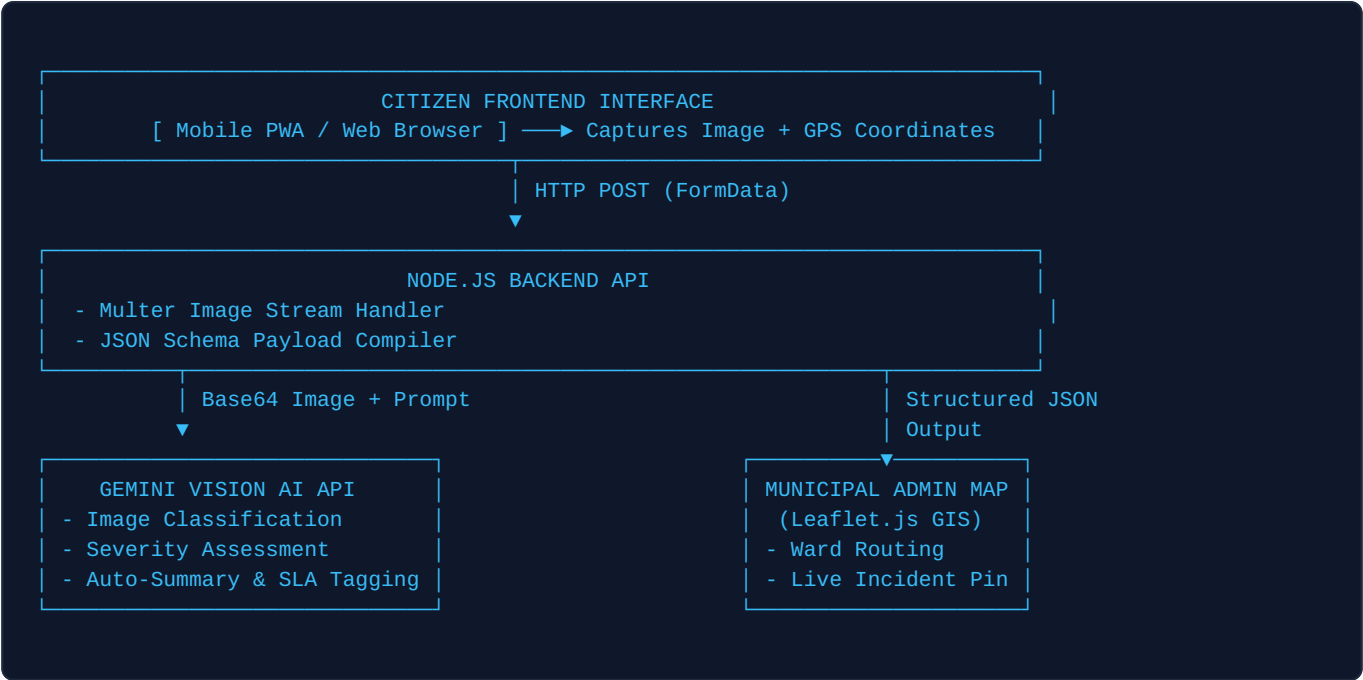
CivicPulse bridges the gap between citizens and municipal bodies through a **Dual-Interface Web Platform**:

Key Value Propositions

Feature	Traditional Process	CivicPulse AI Approach
Reporting Effort	20+ manual form fields	Zero-Form: 1-Click photo upload
Category & Priority	User guess / Manual triage	Automated: Vision AI classification & SLA scoring
Duplicate Handling	Multiple redundant tickets created	Spatial Clustering: Auto-groups within 50m radius
Admin Workflow	Spreadsheet / Paper queues	Live Geospatial Heatmap: Real-time ward dispatch

3. System Architecture

The system utilizes a modern, decoupled architecture designed for low latency, zero user friction, and seamless API integration.



Component Breakdown

- **Frontend Layer:** Responsive HTML5, Tailwind CSS, and Leaflet.js for high-performance GIS map rendering without heavy framework overhead.
- **AI Layer (Gemini 2.5 Flash):** Processes multimodal input (image + prompt) returning structured JSON containing issue category, severity grade, automated summary, and estimated resolution time (SLA).
- **Backend Middleware (Express/Node.js):** Receives raw multipart form requests, handles base64 transformation, executes AI model calls, and updates the spatial database.

4. Features & Implementation Plan

Core Features

- **FEATURE 1 Zero-Form Citizen Reporting:** Automatic geolocation extraction with standard browser Web APIs.
- **FEATURE 2 Multimodal Vision Triage:** Instant image recognition for Potholes, Garbage Accumulation, Water Leaks, and Street Light Failures.
- **FEATURE 3 Automated SLA & Priority Tagging:** Severity calculated dynamically (e.g., Critical vs. Low) based on safety hazard risk.
- **FEATURE 4 Municipal Heatmap & Dispatch Portal:** GIS-based dashboard enabling ward managers to view live cluster maps and update ticket status.

3-Phase Execution Roadmap

Phase	Milestone	Deliverables	Status
Phase 1 (Hackathon MVP)	Core Prototype & AI Engine	Frontend UI, Node backend integration, Gemini Vision processing, Live Leaflet Map pin drop.	COMPLETED
Phase 2 (Pilot Deployment)	Backend Database & Clustering	PostgreSQL/PostGIS integration, auto-deduplication within 50m radius, automated SMS updates.	PLANNED
Phase 3 (Scale & Analytics)	Predictive Analytics & Smart City Integration	Predictive maintenance algorithms for recurring infrastructure damage, multi-language support.	FUTURE